

ORF 445 High Frequency Markets

Homework 3

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March 23, 2026—due midnight Sunday, April 5

In class we talked about several parameters that can be used to characterize different traded assets, and how these are correlated with each other.

- a. Write queries to compute the following quantities for the most active symbol for each instrument in the vector `insts`, across at least one week of data:
 - (i) Average trade size. (This is an easy query against the `trade` table.)
 - (ii) Average quote size (bid size plus ask size), sampled at trade times. You can easily do this with an `aj[]`.
 - (iii) Fraction of trades that occur when the bid-ask spread is larger than one tick. We ignore size: this is the fraction of the number of trades, not the fraction of traded size. “One tick” means the value of `minpxincr` from the `instinfo` table.

Make a plot like the upper panel of Figure 1.

- b. Now compute a third statistic for the most active symbol for each instrument:
 - (iv) Reversion parameter $\eta = \text{Continuations} / \text{Reversals}$.

To do this,

- (1) Extract trades whose price is different than the previous trade price, by `select from (select from trade ...) where differ prc`
- (2) Compute N_C , the number of pairs of consecutive trade price changes in the *same* direction (“continuations”), and N_R , the number of pairs of consecutive trade price changes in the *reverse* direction (“reversals”). Then $\eta = N_C / N_R$. (We ignore the *size* of the price changes, and consider only their direction.)

Make a plot like the lower panel of Figure 1.

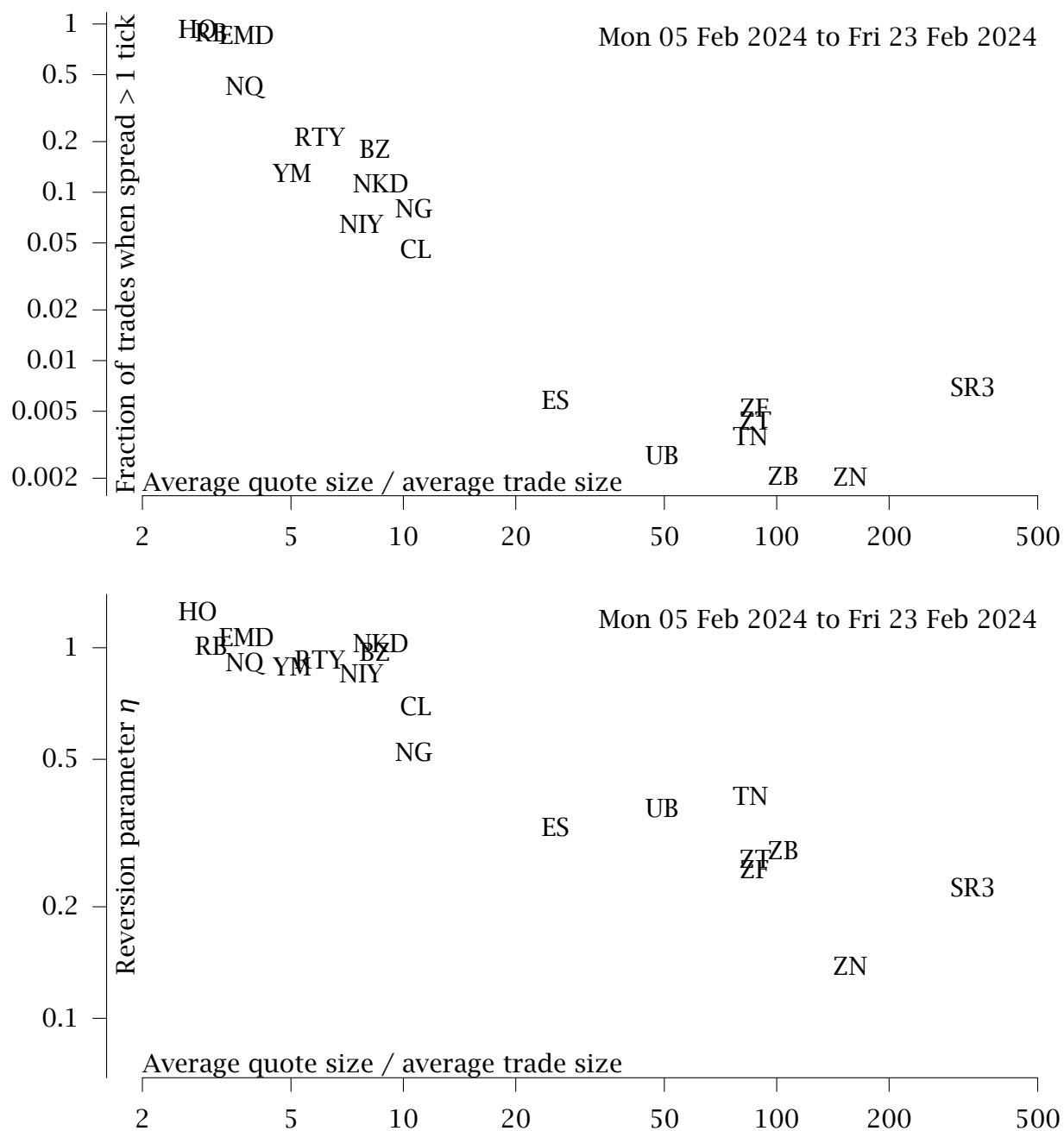


Figure 1: Spectrum of tick sizes. Horizontal axis for both plots is average quote size, sampled at trade times, divided by average trade size. In the upper panel, vertical axis is the fraction (not percent) of trades that occur when the bid-ask spread is greater than the minimum price increment. In the lower panel, vertical axis is the reversion parameter η , the ratio of continuations to reversals in trade prices.

- c. Volatility and correlation. In this problem you will produce a “signature plot” and verify that for large-tick assets, volatility measured on time scale Δt is a decreasing function of Δt . You will also verify the “Epps effect:” that high-frequency correlation in price changes across products goes to zero as the time interval decreases.

Sample bid-ask midpoint prices at one-second intervals (use `til` and `aj[]` to get the last quote in effect at each second interval, even if there were no quote updates in that second), and take differences at a series of different lags. That is, if $p_0, \dots, p_n$ are $n + 1$ prices in one day, then at lag k look at the $n - k + 1$ overlapping differences $p_k - p_0, p_{k+1} - p_1, \dots, p_n - p_{n-k}$, for $k = 1$ up to some reasonable value like 30 or 60. For volatility take the square root of the mean square of these differences divided by the time lag; for correlation take the correlation between these differences.

Sample the midpoint prices at regular time intervals rather than at trade times, since volatility and correlation are intrinsically defined in terms of time rather than trades. Use only a time period during the day, say the 8 hours from 07:00 to 15:00.

- (i) Write an R program with this description:

```
tickvol <- function(
  db,          # handle to historical database
  syms,        # Globex symbol or symbols
  dmin, dmax,  # min and max date
  tmin, tmax,  # min and max times, hours after midnight
  dtmax=60)    # max time difference in seconds
```

It should compute volatility on time interval k , for $k = 1, \dots, dtmax$, where the points are sampled at one-second intervals, using data on the date interval `dmin` to `dmax`, and the time interval `tmin` to `tmax` on each day. For example,

```
tickvol( db, c('ZTH4', 'ZFH4', 'ZNH4', 'ZBH4'),
          '2024-02-05', '2024-02-23', 7, 15 )
```

should produce the upper panel of Figure 2. Why are the volatilities of these interest rate products ordered as $ZT < ZF < ZN < ZB$?

- (ii) Write an R program `tickcorr` like the above, but taking two symbols `sym1` and `sym2` to compute correlation on successive time lags. For example,

```
tickcorr( db, 'ZFH4', 'ZNH4', '2024-02-05', '2024-02-23', 7, 15 )
```

should produce the lower panel of Figure 2. (I have taken only the first two weeks of February for the energy products, because the March contracts expire in the middle of Feb.) The decrease to zero as lag decreases is the Epps effect.

- (iii) Run your codes for Crude Oil and Heating Oil to produce graphs like Figure 3. Why are these products correlated? What is different about Heating Oil from Treasury futures that volatility does not increase on short time horizons? (Hint: compare problems 1 and 2.)

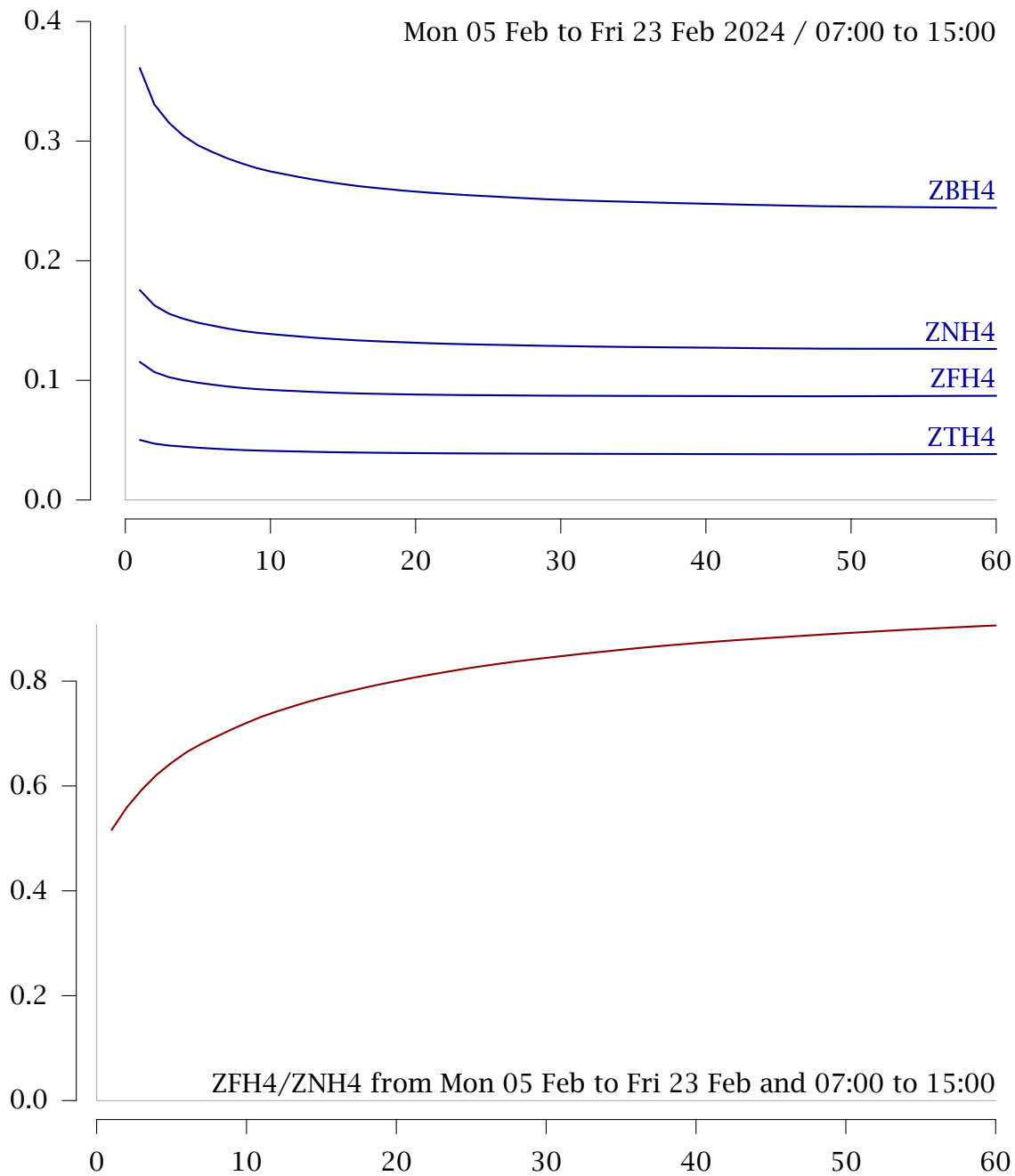


Figure 2: Tick volatility and correlation, for Treasury futures. Horizontal axis is time lag k in seconds. In upper panel, vertical axis is volatility, in native price units, per hour^{1/2}. In lower panel, vertical axis is correlation between ZFH4 and ZNH4 (nondimensional).

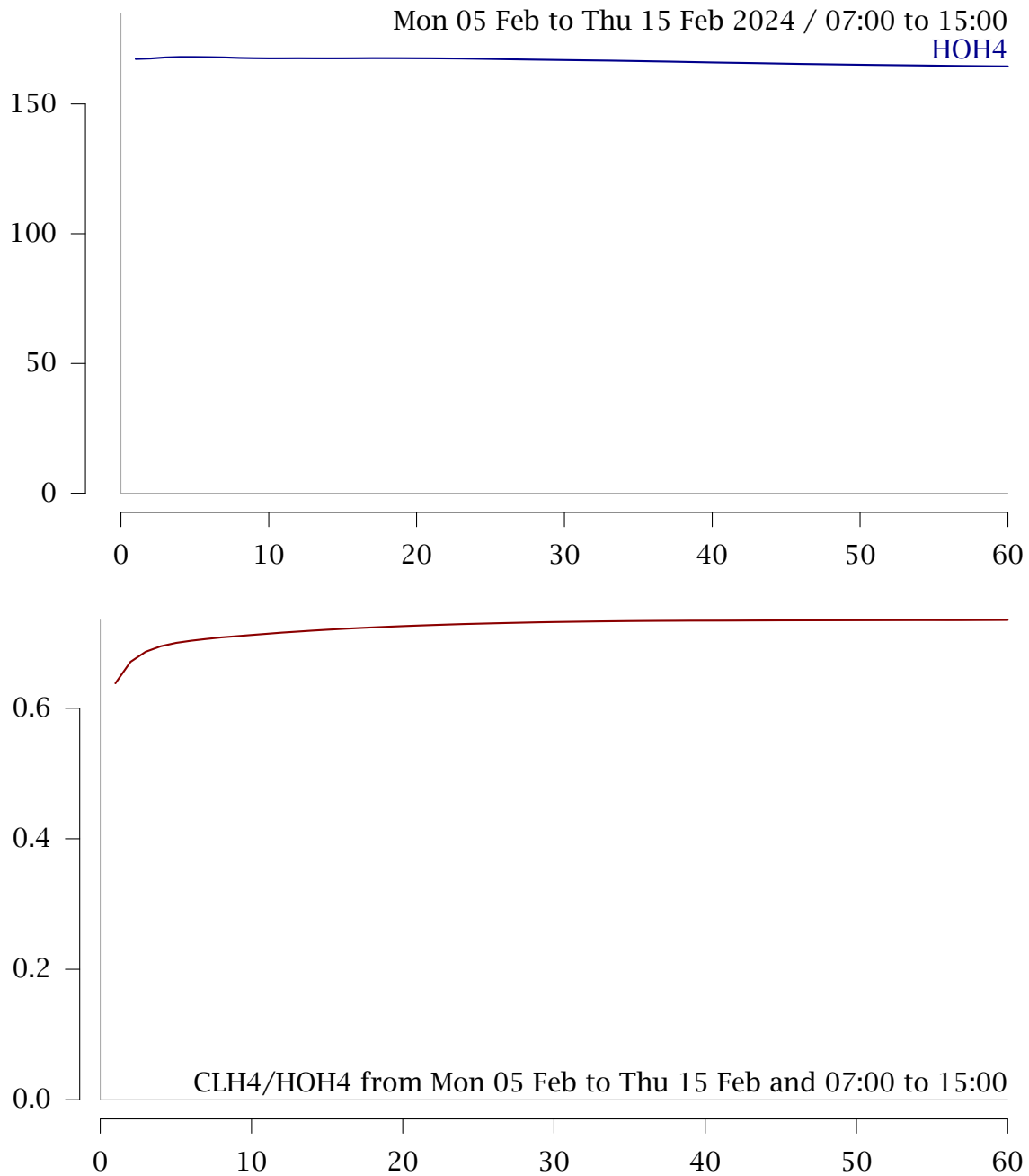


Figure 3: Tick volatility and correlation, like Figure 2 but for Crude Oil and Heating Oil.